

1 • Introduction to probability and probability spaces

Data Analytics for AI • Executive M.Tech • IIT Jammu

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Worked student deck • Interactive probability lab • Rigorous supplement

The slide sequence develops these topics gradually through inspection, measurement, and sampling problems. It may continue across meetings. The main notes serve as an accompanying reference; the rigorous supplement provides the formal details.

Learning objectives

This lecture introduces sets, functions and inverse images, probability models, the probability triplet $(\Omega, \mathcal{F}, P)$, and calculations in finite, countable and continuous spaces. It also reviews the accounts of probability that preceded the axioms, explains why the ratio of favourable to possible cases is a consequence of a modelling assumption rather than a definition, and states why the event collection cannot in general be the set of all subsets. Bertrand's paradox illustrates how a probability depends on the sampling rule. The worked examples require explicit identification of the sample space, events and probability measure.

The mathematical reference is Krishna Jagannathan, *Probability Foundations for Electrical Engineers*, lecture notes, 9 July 2025: Chapter 4, especially §§4.1–4.5, Chapters 1 and 5–6, and Chapter 11 (Definitions 68–70). Chapter 7 provides optional background. The examples below are mathematical illustrations prepared for this course; their numerical values are stipulated assumptions rather than empirical findings. The main notes give the sampling maps and inverse-image calculations. The separate supplement develops the measurable spaces, inverse formulas and distributional derivations in full. The construction theorems of measure theory are outside the scope of this lecture.

1. Probability and statistics in model evaluation

A deterministic classifier produces a fixed output for a given input. Its measured performance may nevertheless vary with the evaluation sample. Probability models describe variation in inputs, labels, sensor noise and training data.

Suppose a quality-inspection system makes 8 errors on a sample of 100 items. The observed **error fraction** is $8/100 = 0.08$. The error fraction on a subsequent sample may differ. Inference about deployment performance requires assumptions about the sampling process and its relation to the deployment population.

Probability calculations determine event probabilities under a specified model. If errors on different items are independent and each has probability p , the probability of no errors in 100 items is $(1 - p)^{100}$. Both the common error probability and independence are model assumptions.

Statistical inference uses observed data to estimate model parameters and assess uncertainty. Under the stated model, the sample error fraction is an estimate of p . Methods for assessing its uncertainty and examining the model assumptions will be developed later.

A probability model may represent physical variability or uncertainty arising from incomplete information. Its suitability depends on the application and the assumptions made.

Example 1: accuracy and class proportions

In a hypothetical collection of 10,000 items, 9,900 are nondefective and 100 are defective. A system that always says "nondefective" has accuracy

$$\frac{9900}{10000} = 0.99,$$

and detects none of the defective items. In this example, overall accuracy depends on the proportion of nondefective items and does not describe detection of defective items. A performance statement therefore needs a specified population, sampling procedure and event of interest.

2. Sets, functions and inverse images

Elements, subsets and collections of subsets

A set is a collection of elements. The notation $\omega \in \Omega$ states that ω is an element of Ω ; $A \subseteq \Omega$ states that every element of A belongs to Ω . The empty set $\emptyset$ has no elements. A singleton $\{\omega\}$ contains one element.

For a die, let $\Omega = \{1, 2, 3, 4, 5, 6\}$ and $A = \{2, 4, 6\}$. Then $2 \in A$ and $A \subseteq \Omega$. The objects 2 , $\{2\}$ and $\{\{2\}\}$ are different: an element, a singleton set, and a collection containing that singleton set.

The **power set** 2^Ω is the collection of all subsets of Ω . For $\Omega = \{H, T\}$,

$$2^\Omega = \{\emptyset, \{H\}, \{T\}, \{H, T\}\}.$$

A probability model selects an event collection $\mathcal{F} \subseteq 2^\Omega$. Thus an event satisfies both $A \subseteq \Omega$ and $A \in \mathcal{F}$.

Set operations

For subsets $A, B \subseteq \Omega$,

$$A \cup B = \{\omega: \omega \in A \text{ or } \omega \in B\},$$

$$A \cap B = \{\omega: \omega \in A \text{ and } \omega \in B\},$$

$$A^c = \Omega \setminus A = \{\omega \in \Omega: \omega \notin A\},$$

$$A \setminus B = \{\omega \in A: \omega \notin B\}.$$

The union uses inclusive "or." Sets are disjoint when their intersection is empty. For $A = \{2, 4, 6\}$ and $B = \{4, 5, 6\}$,

$$A \cup B = \{2, 4, 5, 6\}, \quad A \cap B = \{4, 6\}, \quad A^c = \{1, 3, 5\}.$$

The **Cartesian product** $S \times T = \{(s, t): s \in S, t \in T\}$ is a set of ordered pairs. For two coin tosses, $\{H, T\}^2 = \{HH, HT, TH, TT\}$. For two continuous sampling coordinates, a product such as $(0, 1) \times [0, 1)$ specifies their joint outcome space. Specifying a Cartesian product alone does not assert independence or any probability distribution.

Functions, inverses and inverse images

A function $f: S \rightarrow T$ assigns exactly one value $f(s) \in T$ to each $s \in S$. Its image on a subset $A \subseteq S$ is $f(A) = \{f(s): s \in A\}$.

An **inverse function** $f^{-1}: T \rightarrow S$ exists when f is a bijection: one-to-one and onto the stated codomain. It recovers the original input from the output.

For any function, whether invertible or not, the **inverse image** of a set $B \subseteq T$ is

$$f^{-1}(B) = \{s \in S: f(s) \in B\}.$$

Here $f^{-1}(B)$ is a set of inputs, not a single recovered input. For the head-count function on two tosses, $X(HH) = 2$, $X(HT) = X(TH) = 1$, $X(TT) = 0$.

This function is not one-to-one, so it has no inverse function on $\{0, 1, 2\}$. Its inverse image is nevertheless well defined:

$$X^{-1}(\{1\}) = \{HT, TH\}, \quad X^{-1}([1, 2]) = \{HH, HT, TH\}.$$

Inverse images preserve unions, intersections and complements:

$$f^{-1}(B \cup C) = f^{-1}(B) \cup f^{-1}(C), \quad f^{-1}(B^c) = (f^{-1}(B))^c.$$

The complements are taken in the respective codomain and domain. These identities also hold for countable unions and intersections.

3. The probability triplet

A **probability space** is a triple

$$(\Omega, \mathcal{F}, P).$$

Symbol	Definition	Interpretation
Ω	Sample space	Set of possible outcomes
$\mathcal{F}$	A σ -algebra of subsets of Ω	Events to which probabilities are assigned
P	A probability measure on $\mathcal{F}$	Assignment of a probability to each event

An **outcome** ω is an element of Ω . An **event** A is an element of $\mathcal{F}$ and is itself a subset of Ω . The event occurs exactly when $\omega \in A$. The outcome ω and its singleton event $\{\omega\}$ are distinct mathematical objects. The function P is defined on events.

Example 2: a coin toss

Let $\Omega = \{H, T\}$ and $\mathcal{F} = 2^\Omega = \{\emptyset, \{H\}, \{T\}, \Omega\}$. Here 2^Ω means the **power set**, the collection of every subset of Ω .

Choose a parameter $p \in [0, 1]$ and define

$$P(\{H\}) = p, \quad P(\{T\}) = 1 - p,$$

with $P(\emptyset) = 0$ and $P(\Omega) = 1$. The fair-coin model is the special case $p = 1/2$. The sample space itself imposes no equality on the two probabilities.

Example 3: events for a fair die

For a fair six-sided die, take $\Omega = \{1, 2, 3, 4, 5, 6\}$, $\mathcal{F} = 2^\Omega$ and $P(A) = |A|/6$. Define

$$A = \{2, 4, 6\} \quad (\text{even}), \quad B = \{4, 5, 6\} \quad (\text{greater than } 3).$$

Then $A \cap B = \{4, 6\}$, $A \cup B = \{2, 4, 5, 6\}$ and $A^c = \{1, 3, 5\}$. Their probabilities are $1/3$, $2/3$ and $1/2$ respectively. "Or" means inclusive or unless explicitly stated otherwise.

4. Event collections and σ -algebras

An event can be interpreted as a yes/no question about an outcome. Taking its complement corresponds to negating that question. Taking a union corresponds to asking whether at least one of the constituent events occurs. A σ -algebra provides closure under these operations, including countable unions.

Formally, a collection $\mathcal{F} \subseteq 2^\Omega$ is a **σ -algebra** if:

1. $\Omega \in \mathcal{F}$.
2. If $A \in \mathcal{F}$, then $A^c = \Omega \setminus A \in \mathcal{F}$.
3. If $A_1, A_2, \dots \in \mathcal{F}$, then $\bigcup_{n=1}^{\infty} A_n \in \mathcal{F}$.

These rules also give $\emptyset \in \mathcal{F}$ and closure under countable intersections, because

$$\bigcap_{n=1}^{\infty} A_n = \left(\bigcup_{n=1}^{\infty} A_n^c \right)^c.$$

The word **countable** includes finite collections and collections that can be indexed by $1, 2, 3, \dots$. It does not mean an arbitrary collection of any size.

Example 4: a die observed only through parity

Suppose a device reports only "odd" or "even." We retain the underlying outcomes $\Omega = \{1, \dots, 6\}$ but describe the reported information using

$$\mathcal{F}_{\text{parity}} = \{\emptyset, \{1, 3, 5\}, \{2, 4, 6\}, \Omega\}.$$

This is a σ -algebra. Complements exchange the two parity sets, and any union of these four sets is still one of the four. The set $\{6\}$ does not belong to this σ -algebra, so $P(\{6\})$ is undefined in this model. A model with $\mathcal{F} = 2^\Omega$ includes this singleton as an event.

In this example, the σ -algebra represents the information available from the parity report. More generally, it specifies the domain of the probability measure and need not describe a physical observation device.

Counterexample. The collection $\{\emptyset, \Omega, \{1\}\}$ is not a σ -algebra on the die space: it omits the complement $\{2, 3, 4, 5, 6\}$.

Countable unions

Let A_n mean "the first successful transmission occurs at attempt n ." The question "does a success ever occur?" is $\bigcup_{n=1}^{\infty} A_n$. Closure under finite unions alone does not guarantee that this event is included.

A precise counterexample uses $\Omega = \mathbb{N}$ and the algebra of finite or cofinite subsets. It contains every singleton but not the set of positive even integers, although that set is a countable union of singletons. This example shows that closure under finite unions does not imply closure under countable unions. An algebra that also has countable-union closure is a σ -algebra.

For finite or countable Ω , the full power set can be used as the event σ -algebra. Standard continuous length and area models use σ -algebras such as the Borel sets. Their events include the intervals and disks used in this lecture.

Sets to which no probability can be assigned

A natural question is why $\mathcal{F}$ is not simply taken to be 2^Ω in every model. For finite and countable sample spaces it can be. For the uniform model on an interval it cannot, and the obstruction is a theorem rather than a limitation of present technique.

Consider $\Omega = [0, 1)$ with addition modulo 1, written $x \oplus y$. This operation moves a point along the interval and wraps it around, so it is a rotation of the circle obtained by joining the two endpoints. The uniform model on Ω is expected to satisfy three conditions:

1. $P([0, 1)) = 1$;
2. P is countably additive;
3. P is **rotation invariant**: $P(A \oplus t) = P(A)$ for every event A and every t , where $A \oplus t = \{a \oplus t : a \in A\}$.

The third condition states that a rigid shift does not change length. It is the property that makes the model uniform.

Theorem (Vitali, 1905). There is no function $P: 2^{[0,1)} \rightarrow [0, 1]$ satisfying conditions 1–3.

The proof exhibits a set $V \subseteq [0, 1)$ to which no value can be assigned. Call x and y equivalent when $x - y$ is rational. This relation partitions $[0, 1)$ into classes, and V is formed by selecting exactly one member of each class. The rational translates $V \oplus q$, for q ranging over the countably many rationals in $[0, 1)$, are then pairwise disjoint and together exhaust $[0, 1)$. Rotation invariance would give every translate the same probability c ; countable additivity would then give $1 = \sum_q c$, which is 0 when $c = 0$ and unbounded when $c > 0$. Neither value is admissible. The construction is given in full in the supplement.

Two remarks are needed to read this correctly.

The selection uses the axiom of choice. No rule is given for picking a representative from each class; the axiom asserts only that a selection exists. Solovay (1970) showed that it is consistent with Zermelo–Fraenkel set theory together with dependent choice that every set of reals is Lebesgue measurable, so a non-measurable set cannot be exhibited explicitly. No set arising from a description of a measurement is of this kind.

The consequence is the choice of domain, not a defect in the model. The uniform model on $[0, 1)$ exists and is entirely standard once $\mathcal{F}$ is taken to be the Borel σ -algebra, or its completion. Every event used in this lecture — intervals, disks, angular sectors, and the countable combinations of them — is a Borel set. The restriction of $\mathcal{F}$ is what allows conditions 1–3 to hold simultaneously.

Two distinct reasons for restricting the event collection

The parity example and the Vitali theorem both limit $\mathcal{F}$, and they do so for different reasons. The distinction is worth keeping.

	Source of the restriction	Under the modeller's control
Vitali	Mathematical necessity: conditions 1–3 are jointly unsatisfiable on $2^{[0,1)}$	No
Parity report	Modelling decision: $\mathcal{F}$ records which questions the measurement answers	Yes

Only the second is a statement about an experiment. The first is a statement about sets.

Information that a measurement does not record cannot be recovered

The second kind of restriction has a consequence that matters in applications. Suppose the recorded quantity is Y , and $\mathcal{G} = \sigma(Y)$ is the σ -algebra of questions its value answers. Let g be any measurable function and consider the processed record $g(Y)$. Then

$$\sigma(g(Y)) \subseteq \sigma(Y) = \mathcal{G}.$$

The inclusion holds because $\{g(Y) \in B\} = \{Y \in g^{-1}(B)\}$, which is already an element of $\mathcal{G}$. Every question answerable from the processed record was therefore answerable from the original record. No computation applied after the measurement enlarges the collection of events, and the inclusion is usually strict: a non-injective g discards distinctions permanently.

In the parity model this says that no function of the reported parity determines whether the face was 6. The report takes the same value on the outcomes 2, 4 and 6, so any function of it takes the same value on all three.

Example 4b: an alert log recorded to the nearest hour

A monitoring system records the hour in which each alert occurred but not the time within the hour. Take $\Omega = [0, 24)$ in hours and let $\mathcal{G}$ be the σ -algebra generated by the 24 intervals $[k, k + 1)$. The event “an alert occurred between 09:00 and 10:00” is $[9, 10) \in \mathcal{G}$ and has a probability in this model. The event

“an alert occurred in the first minute of that hour” is $[9, 9 + \frac{1}{60})$, which is not in $\mathcal{G}$ and is not assigned a probability.

The preceding inclusion explains what this costs. Because every statistic computed from the log is $\mathcal{G}$ -measurable, no reanalysis of the stored data answers the minute-level question. Obtaining an answer requires changing the measurement — recording finer timestamps — not changing the calculation. Choosing $\mathcal{F}$ is therefore part of specifying the experiment, and it is fixed at the point of measurement rather than at the point of analysis.

5. Interpretations of probability before the axioms

The triplet and the axioms that follow were set out by Kolmogorov in *Grundbegriffe der Wahrscheinlichkeitsrechnung* (1933). They fix the calculus; they do not state what a probability means. Several accounts of the meaning were in use before 1933 and remain in use now. This section describes them, explains why the counting rule that opens most elementary treatments is not a definition, and identifies what the measure-theoretic treatment supplied that the earlier accounts did not.

5.1 The classical definition

Laplace, in *Théorie analytique des probabilités* (1812), defined the probability of an event as the number of favourable cases divided by the number of equally possible cases:

$$P(A) = \frac{\text{number of favourable cases}}{\text{number of equally possible cases}}.$$

Cases are taken to be equally possible when there is no reason to prefer one to another, a rule Keynes (1921) later named the **principle of indifference**. The definition is adequate for the symmetric games from which the subject arose, and it gives the right answers for a fair die.

Four difficulties prevent it from serving as a foundation.

It is circular. “Equally possible” is the notion being defined. A statement that two cases are equally possible is already a probabilistic assertion, so the definition explains probability in terms of a particular probability.

It depends on the chosen partition. The same experiment admits several exhaustive lists of cases, and they need not agree. In the *Encyclopédie* article “Croix ou Pile” (1754), d’Alembert considered obtaining at least one head in two tosses of a coin and listed three cases: a head at once; a tail then a head; two tails. Counting gives $2/3$. The standard model of two independent fair tosses gives $3/4$. The list is not wrong as a sample space — exactly one of the three occurs — but its members carry probabilities $1/2$, $1/4$ and $1/4$, not $1/3$ each. Counting is valid only for a partition whose members are in fact equally probable, and the definition provides no way to check that.

It does not extend to countably infinite sample spaces. Suppose $\Omega = \{1, 2, 3, \dots\}$ and every outcome is assigned the same probability c . Countable additivity gives

$$1 = P(\Omega) = \sum_{n=1}^{\infty} c,$$

which is 0 when $c = 0$ and unbounded when $c > 0$. No uniform distribution on a countably infinite set exists, so there is no ratio to form.

It does not extend to continuous sample spaces. When Ω is an interval or a disk, both counts are infinite and the ratio is undefined. Replacing counting by length, area or angle restores a ratio, but the choice among those quantities is exactly what has to be justified. Bertrand (1889) showed that three defensible choices give three different answers to one geometric question; that construction is developed in Section 8.

The modern position is that the counting rule is a **consequence of a modelling assumption rather than a definition**. If Ω is finite and P is the uniform measure, then $P(A) = |A|/|\Omega|$ follows from the axioms. Example 3 uses it on that basis, and Example 5 shows a die for which it does not apply. Which model holds is an assumption about the experiment, and it is not supplied by the mathematics.

The empirical form of the same ratio — the number of trials in which A occurred divided by the number of trials — is a different quantity. It is an observed statistic, not a probability, and the relation between the two is the subject of the next interpretation and of the limit theorems.

5.2 The frequency interpretation

Venn (*The Logic of Chance*, 1866) and von Mises (1919; *Wahrscheinlichkeit, Statistik und Wahrheit*, 1928) identified the probability of A with the limiting relative frequency of A in an indefinitely long sequence of repetitions. Von Mises made this precise through the notion of a **collective**: an infinite sequence of outcomes in which the relative frequency of each attribute converges, and in which the limit is unchanged by any admissible rule for selecting a subsequence without reference to the outcomes selected.

The interpretation matches the way probabilities are checked against data and underlies most reporting of error rates. Three difficulties are usually raised.

The existence of the limit is postulated rather than derived, and no finite record establishes it. The class of admissible selection rules must be restricted, since permitting every subsequence makes the requirement unsatisfiable; later work by Church and by Wald restricted it to computable rules. A single event that is not part of a repeatable sequence receives no probability.

There is a further consequence specific to this course. Under the frequency definition, the convergence of relative frequencies is true by construction. Under the axioms it is a theorem — Borel (1909) proved such a result for the binary digits of a number chosen uniformly, and the strong law of large numbers is its general form. A statement that can be proved is more useful than one that has been assumed, because its hypotheses are then visible and can be checked.

5.3 Degrees of belief

A second family of accounts treats probability as a degree of belief rather than a property of a repeatable sequence.

Ramsey (1926) and de Finetti (1931, 1937) developed the **subjective** account, in which a person's probability for A is the betting quotient they will accept on A . The requirement imposed is **coherence**: the quotients must not permit a set of bets that loses in every outcome. Coherence forces the quotients to obey the finite addition rules, which is how the calculus is recovered from the interpretation rather than assumed. De Finetti held that only finite additivity is justified on this

reasoning, which is a genuine mathematical divergence from Kolmogorov's third axiom and not a difference of emphasis: several standard limit results fail without countable additivity. His representation theorem for exchangeable sequences (1937) supplies, for this account, the role played by independent repetition in the frequency account.

Keynes (*A Treatise on Probability*, 1921), Jeffreys (*Theory of Probability*, 1939) and Carnap (*Logical Foundations of Probability*, 1950) developed the **logical** account, in which probability is a relation between a body of evidence and a proposition, analogous to partial entailment and not dependent on any individual's opinion. The programme requires a canonical assignment in the absence of evidence, and the principle of indifference inherits here the partition-dependence described in Section 5.1.

5.4 Fiducial probability

Fisher (1930) proposed the **fiducial** argument, which seeks probability statements about an unknown parameter using the sampling distribution alone, without assuming a prior distribution for the parameter. The aim was to obtain the interval statements of applied statistics without the subjective element of the preceding section.

Two points should be noted, as the term is still encountered.

It is an interpretation of **inferential statements**, not an account of the meaning of the probability calculus, and so it is not an alternative to the axioms in the sense that Sections 5.2 and 5.3 are. It is also nearly contemporaneous with the axioms rather than clearly prior to them.

The argument does not extend reliably beyond the simplest cases. In problems with more than one unknown parameter the resulting distributions are not unique, different routes of derivation can disagree, and the intervals need not have the stated coverage; the Behrens–Fisher problem is the standard illustration. The idea has descendants in current work on confidence distributions and generalized fiducial inference, but the original argument is not a general method.

The dispute matters for this course because it is the dispute about what a reported interval means. That question returns when model performance is estimated from a finite evaluation sample.

5.5 Why Bertrand's paradox is only part of the explanation

Bertrand's paradox is often presented as the reason the subject was rebuilt on measure theory. It is a genuine motivation, and Section 8 develops it in full, but it does not account for what Kolmogorov actually did.

What the paradox establishes is that the classical rule is **ill-posed on a continuum**: "choose a chord at random" does not determine a probability until the mechanism generating the chord is specified. Both the diagnosis and the remedy can be stated in classical language. The remedy is a discipline of modelling — say which quantity is uniform — and it requires no new mathematics. Bertrand himself presented the example as showing that the question had been badly posed.

What the paradox does not supply is a framework in which statements about infinite sequences and continuous-time processes can be formulated and proved. Four developments account for that.

Measure theory already existed and already worked. Borel (1898) and Lebesgue (1902) had constructed a theory of length and integration for a wide class of sets. Borel (1909) then used it to prove a result about the digits of a randomly chosen number that is recognisably a probability theorem.

The tools were therefore available, and had already produced results the older frameworks could not state.

Limit theorems are assertions about infinite sequences. "The proportion of errors converges to p " is a statement about an infinite sequence of trials. Before it can be given probability 1, the set of sequences on which the convergence holds must be an event. Constructing a probability measure on such a space of infinite sequences, and making the convergence set belong to the σ -algebra, is what countable additivity and the closure requirement of Section 4 are for. Finite additivity is insufficient.

Processes indexed by a continuum required measures on function spaces. Wiener (1923) constructed Brownian motion as a measure on a space of continuous functions. Kolmogorov's extension theorem states conditions under which a consistent family of finite-dimensional distributions determines a single measure on an infinite product, which is the general form of that construction.

Conditioning on an event of probability zero required a new definition. The elementary formula $P(A | B) = P(A \cap B)/P(B)$ is undefined when $P(B) = 0$, and in a continuous model this is the ordinary situation: conditioning on an exact measured value conditions on a null event. Kolmogorov defined conditional expectation relative to a σ -algebra, determined only up to sets of probability zero. The associated Borel–Kolmogorov paradox shows that conditioning on a great circle of a sphere has no answer until the family of conditioning events is specified. This is Bertrand's lesson again in a setting where specifying the mechanism is not merely good practice but part of the definition, and it is a genuinely new object that no discipline of modelling produces on its own.

Hilbert's sixth problem (1900) had also asked for an axiomatic treatment of probability alongside mechanics, and Kolmogorov's monograph is a response to that request.

The summary is that Bertrand's paradox shows a probability model must be **specified**, while the measure-theoretic treatment supplies a **language in which models can be specified and limit theorems proved**. These are different achievements, and only the second required the new mathematics.

5.6 What the axioms settle

The axioms in the next section fix the rules of calculation and nothing else. All the interpretations above use the same rules; they differ over what the resulting numbers assert and over how a particular P is to be chosen. The choice of Ω , of $\mathcal{F}$ and of P is a modelling decision and lies outside the mathematics, which is why Section 1 begins with the assumptions and Section 8 concerns a question that has three answers.

6. Kolmogorov's axioms

Once the events are specified, $P: \mathcal{F} \rightarrow [0, 1]$ must satisfy:

1. **Nonnegativity:** $P(A) \geq 0$ for every $A \in \mathcal{F}$.
2. **Normalization:** $P(\Omega) = 1$.
3. **Countable additivity:** for pairwise disjoint $A_1, A_2, \dots \in \mathcal{F}$,

$$P\left(\bigcup_{n=1}^{\infty} A_n\right) = \sum_{n=1}^{\infty} P(A_n).$$

Pairwise disjoint means $A_i \cap A_j = \emptyset$ whenever $i \neq j$. The additivity axiom applies to disjoint events.

Elementary consequences

Empty event. Apply countable additivity to $\Omega, \emptyset, \emptyset, \dots$. Nonnegative probabilities and the finite left side force $P(\emptyset) = 0$. Finite additivity follows by appending empty events.

Complement. Since A and A^c form a disjoint partition of Ω ,

$$1 = P(A) + P(A^c), \quad P(A^c) = 1 - P(A).$$

Monotonicity. If $A \subseteq B$, then $B = A \dot{\cup} (B \setminus A)$, so $P(B) = P(A) + P(B \setminus A) \geq P(A)$. The dot stresses a disjoint union.

Two-event union. Partition $A \cup B$ into A and $B \setminus A$. Also partition B into $B \setminus A$ and $A \cap B$. Eliminating $P(B \setminus A)$ gives

$$P(A \cup B) = P(A) + P(B) - P(A \cap B).$$

As a consequence, $P(A \cup B) \leq P(A) + P(B)$, the two-event union bound.

7. Random variables and probabilities of numerical events

A real-valued random variable is a measurable function $X: \Omega \rightarrow \mathbb{R}$. The name can mislead: X is a fixed rule once the experiment and recording rule are specified; the outcome ω is random, and $X(\omega)$ is its numerical record. The measurability condition requires

$$X^{-1}(B) = \{\omega: X(\omega) \in B\} \in \mathcal{F}$$

for every Borel set B of real values. Intervals and singleton sets are sufficient for the examples here; the formal definition is developed in the supplement.

A random variable is a function on the set of **outcomes**. The probability measure $P: \mathcal{F} \rightarrow [0, 1]$ is a **set function**: it assigns a number in the unit interval to each event. The inverse-image map connects these two types of object:

$$B \mapsto X^{-1}(B) \mapsto P(X^{-1}(B)).$$

The distribution of X is the set function $P_X(B) = P(X^{-1}(B))$. It assigns a probability to each measurable set of numerical values.

For two independent fair tosses, let X count the heads. Since $X^{-1}(\{1\}) = \{HT, TH\}$,

$$P(X = 1) = P(\{HT, TH\}) = 1/2.$$

The function X has no inverse function, since both HT and TH map to 1. The inverse image remains well defined.

For an event A , the indicator $\mathbf{1}_A$ equals 1 on A and 0 on A^c . Thus $\mathbf{1}_A^{-1}(\{1\}) = A$ and $P(\mathbf{1}_A = 1) = P(A)$. An indicator is a random variable; the assignment $A \mapsto P(A)$ is a set function.

8. Bertrand's paradox: specification of the three models

Chords, coordinates and the event

Fix a circle of radius R . For a chord c , let $m(c)$ be its midpoint. Use the coordinates

Here Arg is taken in $[0, 2\pi)$.

$$\phi = \text{Arg}(m(c)), \quad t = \|m(c)\|/R.$$

For $0 < t < 1$, the forward map $C(\phi, t)$ places the midpoint at $Rt(\cos\phi, \sin\phi)$ and draws the chord perpendicular to that radius. Its inverse recovers the midpoint angle and normalized distance:

$$C^{-1}(c) = (\text{Arg}(m(c)), \|m(c)\|/R).$$

Take $0 \leq \phi < 2\pi$. Diameters and zero-length chords are excluded from this chart; they have probability zero under all three sampling rules. The supplement gives explicit endpoint formulas and specifies the measurable spaces.

The **chord-length random variable** is

$$X(c) = 2R\sqrt{1 - T(c)^2}, \quad T(c) = \|m(c)\|/R.$$

The side of the inscribed equilateral triangle has length $\sqrt{3} R$. Thus the event being measured is

$$E = \{c: X(c) > \sqrt{3} R\} = \{c: T(c) < 1/2\}.$$

Sampling distributions, maps and inverses

Let U, V be independent uniform random variables on $(0, 1)$ and $[0, 1)$, respectively. Write Q for their joint uniform probability measure. The three sampling maps have the form

$$\Psi_j(u, v) = C(2\pi v, q_j(u)).$$

Model	Sampling rule	$t = q_j(u)$	Inverse $u = q_j^{-1}(t)$
A	Independent uniform endpoints	$\cos(\pi u/2)$	$(2/\pi)\arccost$
B	Uniform radial distance	u	t
C	Uniform midpoint in disk area	$\sqrt{u}$	t^2

In every row, the angular inverse is $v = \phi/(2\pi)$. Consequently, a sampled chord determines its sampling coordinates through

$$\Psi_j^{-1}(c) = \left(q_j^{-1}(T(c)), \frac{\text{Arg}(m(c))}{2\pi} \right).$$

For A, the original endpoint angles are independent and uniform on $[0, 2\pi)$. The half-angle H of the minor arc is uniform on $(0, \pi/2)$, so $U = 2H/\pi$ is uniform and $T = \cos H$. The midpoint direction is independent and uniform. The original map from **ordered** endpoints to an unordered chord is two-to-one. The table uses the equivalent one-to-one midpoint/half-angle parametrization. Its justification and the recovery of the unordered endpoints are in the supplement.

For B, uniform radial distance gives T a uniform distribution on $(0, 1)$; it can be generated by setting $T = U$.

For C, uniform disk area implies $P(T \leq t) = \pi(Rt)^2/(\pi R^2) = t^2$, so T has the same distribution as $\sqrt{U}$. This can be generated by setting $T = \sqrt{U}$. Setting $T = U$ would instead specify Model B.

Inverse images give the three probabilities

The induced chord measure is $P_j(F) = Q(\Psi_j^{-1}(F))$ for every measurable chord event F . The length random variable on the uniform source space is $X_j = X \circ \Psi_j$.

Model	Length $X_j(u, v)$	Event $\Psi_j^{-1}(E)$	Probability
A	$2R\sin(\pi u/2)$	$u > 2/3$	1/3
B	$2R\sqrt{1-u^2}$	$u < 1/2$	1/2
C	$2R\sqrt{1-u}$	$u < 1/4$	1/4

For example, in A,

$$X_A > \sqrt{3} R \Leftrightarrow \sin(\pi U/2) > \sqrt{3}/2 \Leftrightarrow U > 2/3.$$

Since U is uniform, this event has probability $1 - 2/3 = 1/3$. The other two rows follow by solving their inequalities in U .

The same chord event E has different probabilities because P_A, P_B, P_C are different measures. On the common uniform source space, the event preimages $\Psi_j^{-1}(E) = \{(u, v): X_j(u, v) > \sqrt{3} R\}$ are different sets, with Q -measures 1/3, 1/2, and 1/4. A coordinate change within one fixed model preserves event probabilities.

A coordinate change within a fixed model

Keep Model B and put $W = T^2$. Its inverse is $T = \sqrt{W}$. Then

$$P_B(W < 1/4) = P_B(T < 1/2) = 1/2.$$

In this model, $P_B(W \leq w) = \sqrt{w}$, so W is not uniform. Declaring W uniform would change the sampling distribution to Model C. Correct reparametrization preserves probability; changing the distribution need not.

Interactive sampling maps and inverse images · Rigorous derivations

9. Worked examples

Example 5: a die with unequal outcome probabilities

Suppose a loaded die has $P(\{i\}) = i/21$ for $i = 1, \dots, 6$. The weights are nonnegative and sum to $(1 + 2 + 3 + 4 + 5 + 6)/21 = 1$. With the full power set,

$$P(A) = \sum_{i \in A} \frac{i}{21}.$$

For $A = \{2, 4, 6\}$, $P(A) = 12/21 = 4/7$. For $B = \{4, 5, 6\}$, $P(B) = 15/21 = 5/7$. Their intersection has weight $10/21$, so

$$P(A \cup B) = \frac{12 + 15 - 10}{21} = \frac{17}{21}.$$

"Number favourable divided by number possible" works only for a finite model with equally weighted outcomes.

Example 6: two coin tosses and the head count

Assume independent fair tosses. The ordered outcomes HH, HT, TH, TT each have probability $1/4$. Thus

$$P(\text{at least one head}) = 1 - P(\{TT\}) = 3/4.$$

If we record only the number X of heads, the possible values are $\{0, 1, 2\}$, but their probabilities are $1/4, 1/2, 1/4$. The value $X = 1$ corresponds to two outcomes, HT and TH , while each of the other values corresponds to one outcome.

For independent biased tosses with head probability p , the same calculation gives $P(X = 1) = 2p(1 - p)$ and $P(X \geq 1) = 1 - (1 - p)^2$. The multiplication uses independence, an additional model assumption.

Example 7: two fair dice

Assume independent fair dice. Use $\Omega = \{1, \dots, 6\}^2$, its power set, and probability $1/36$ on each ordered pair.

A sum of 7 occurs at $(1, 6), (2, 5), (3, 4), (4, 3), (5, 2), (6, 1)$, hence has probability $6/36 = 1/6$. The event of at least one six is the complement of the event that neither die shows six:

$$P(\text{at least one six}) = 1 - \frac{5 \cdot 5}{36} = \frac{11}{36}.$$

The possible sums $2, 3, \dots, 12$ are not equally likely: sum 2 has one generating pair, while sum 7 has six.

Example 8: overlapping system alerts

In a hypothetical system model, let A be a latency alert and B an error-rate alert. Suppose $P(A) = 0.20$, $P(B) = 0.15$ and $P(A \cap B) = 0.08$.

$$P(A \cup B) = 0.20 + 0.15 - 0.08 = 0.27.$$

The disjoint regions have weights: latency only 0.12, error-rate only 0.07, both 0.08, neither 0.73. They are nonnegative and sum to 1, so they define a valid four-outcome model. No independence assumption was needed.

If a report instead claimed $P(A \cap B) = 0.25$, it would be invalid because an intersection cannot have greater probability than either containing event.

Example 9: a randomly selected record

Consider a fixed dataset containing 80 records from site A and 20 from site B. Select one of the 100 records uniformly. With individual records as outcomes and the full power set as events,

$$P(\text{site B}) = 20/100 = 0.20.$$

Now change the sampling mechanism: select the site by a fair coin, then select a record uniformly within that site. The site-B probability is $1/2$. Individual site-A records have probability $1/160$ and site-B records have probability $1/40$. These weights sum to $80/160 + 20/40 = 1$.

These procedures assign different probability measures to the same set of records. The dependence on the sampling rule is analogous to that in Bertrand's example.

Example 10: a continuous arrival time

Model an arrival time T as uniform on $[0, 10]$ minutes. Take $\Omega = [0, 10]$, $\mathcal{F} = \mathcal{B}([0, 10])$ (the σ -algebra generated by the intervals), and $P(A) = \lambda(A)/10$, where λ denotes length measure on these sets.

$$P(2 \leq T \leq 5) = \frac{5-2}{10} = 0.30, \quad P(T = 3) = 0.$$

A singleton has length measure zero. Thus a nonempty event can have probability zero. Its complement has probability one even though it does not contain every outcome.

The interval is an uncountable union of singletons, whereas the axiom requires additivity for countable disjoint unions. Every countable set of times has probability zero under this continuous model. A reported rounded timestamp such as $Y = \text{round}(T)$ is a discrete random variable obtained by grouping intervals of continuous time.

Example 11: first success in independent trials (extension)

Assume independent attempts, each succeeding with probability p , with $0 < p \leq 1$. Let N be the first successful attempt. Use $\Omega = \{1, 2, \dots\} \cup \{\infty\}$, where ∞ records no success ever, and $\mathcal{F} = 2^\Omega$. Define

$$P(\{n\}) = (1-p)^{n-1}p, \quad n \geq 1.$$

The geometric series sums to 1, so $P(\{\infty\}) = 0$ gives a normalized measure. We include the never-success outcome to distinguish an empty event from a null event.

For $p = 1/2$,

$$P(N \text{ even}) = \sum_{k=1}^{\infty} 2^{-2k} = \frac{1/4}{1-1/4} = \frac{1}{3}.$$

This calculation applies countable additivity to the disjoint singleton events $\{2\}, \{4\}, \dots$. For general p , the same sum is $(1 - p)/(2 - p)$.

Example 12: disjointness and independence (extension)

For a fair die, $A = \{1\}$ and $B = \{2\}$ are disjoint: $P(A \cap B) = 0$. Independence would require $P(A \cap B) = P(A)P(B) = 1/36$, which fails. The two distinct faces cannot occur on the same roll.

By contrast, in the independent fair two-toss model, "first toss is H" and "second toss is H" overlap at HH , yet are independent: $1/4 = (1/2)(1/2)$. Disjointness means an empty intersection. Independence is the equality $P(A \cap B) = P(A)P(B)$ under the specified measure.

10. Exercises and solutions

1. A fair die is rolled. Let $A = \{1, 2, 3\}$ and $B = \{3, 4\}$. Find the probabilities of the intersection, union and neither event.
2. On $\Omega = \{a, b, c\}$, is $\{\emptyset, \Omega, \{a\}, \{b, c\}\}$ a σ -algebra? Is $\{\emptyset, \Omega, \{a\}, \{b\}\}$ one?
3. Determine whether 0.2, 0.3, 0.6 are valid outcome probabilities for a three-outcome space. If the first two probabilities are fixed, determine the third.
4. Two independent fair coins are tossed. Compute the ratio of the probability of at least one head to the probability of two heads.
5. A chord generator chooses D/R uniformly on $[0, 1]$. Another chooses a midpoint uniformly by area. Find $P(D < R/3)$ under each.
6. A time is uniform on $[0, 10]$. Find $P(T \in \{1, 2, 3, \dots, 10\})$.
7. For the alert model of Example 8, calculate the probability of exactly one alert.
8. Write a triplet for a randomly selected item from your workplace. State the sampling assumption explicitly, and give one event and its complement.
9. For the head-count function on two tosses, find $X^{-1}(\{0, 2\})$. Explain why it is defined even though X has no inverse function.
10. A chord has normalized midpoint distance $T = 1/2$ and midpoint angle $\phi = \pi$. Recover (u, v) under each Ψ_j^{-1} . Does it belong to the strict event $X > \sqrt{3} R$?
11. In Model B, write the event $T < 1/3$ in terms of $W = T^2$. Calculate its probability using the distribution function of W .

Solutions

1. $A \cap B = \{3\}$ gives $1/6$; $A \cup B = \{1, 2, 3, 4\}$ gives $2/3$; neither is $\{5, 6\}$ and gives $1/3$.
2. The first is a σ -algebra: its two nontrivial sets are complements and all unions stay inside it. The second is not: it lacks the complement $\{b, c\}$ of $\{a\}$ and also the union $\{a, b\}$.
3. The weights sum to 1.1, violating normalization. The third weight must be $1 - 0.2 - 0.3 = 0.5$.

4. The probabilities are $3/4$ and $1/4$, so their ratio is 3.
5. Uniform radial distance gives $1/3$; uniform area gives $(1/3)^2 = 1/9$.
6. Zero. Each singleton has probability zero and a finite union still has probability zero.
7. $P(A \setminus B) + P(B \setminus A) = 0.12 + 0.07 = 0.19$.
8. One valid example: select one of N logged requests uniformly; let Ω be their identifiers, $\mathcal{F} = 2^\Omega$, and $P(A) = |A|/N$. The event "latency exceeds 200 ms" is a subset of identifiers; its complement is "latency is at most 200 ms." Uniform record selection is an explicit assumption and differs from sampling users uniformly.
9. $X^{-1}(\{0, 2\}) = \{TT, HH\}$. An inverse image collects all inputs whose values belong to the specified set; it does not require injectivity.
10. $v = 1/2$ in all three models. Model A gives $u = 2/3$, Model B gives $u = 1/2$, and Model C gives $u = 1/4$. The chord has length exactly $\sqrt{3} R$, so it is excluded from the strict event in every case.
11. The event is $W < 1/9$. Its probability is $P(W < 1/9) = \sqrt{1/9} = 1/3$. Assigning probability $1/9$ would incorrectly treat W as uniform under Model B.

11. Summary and further reading

A probability space consists of a sample space, a σ -algebra of events and a normalized countably additive measure. Equal outcome probabilities are an assumption of a particular model, so the ratio of favourable to possible cases is a result that holds for the uniform model on a finite sample space and not a definition of probability. The event collection is restricted for two distinct reasons: as a modelling decision recording what a measurement reports, and, for the uniform model on an interval, as a mathematical necessity established by Vitali's theorem. Information a measurement does not record cannot be recovered by later computation, because any measurable function of the record generates a smaller σ -algebra. Random variables map outcomes to numerical values; inverse images convert numerical sets into events. Bertrand's three sampling rules induce different measures on the same chord space. Their maps, inverses and densities specify why the three probabilities differ.

The interpretations in Section 5 are not settled by the axioms and are not in competition with them: the classical, frequency, subjective and logical accounts all use the same calculus and differ over what the numbers assert. Bertrand's paradox motivates specifying a model; the measure-theoretic treatment was required for a different purpose, namely stating and proving limit theorems about infinite sequences, constructing processes indexed by a continuum, and defining conditional probability given an event of probability zero.

The next lecture develops conditional probability and independence. Suggested reading is Jagannathan Chapters 1 and 4, the elementary properties in Chapter 5, and Chapter 11, Definitions 68–70. Measure construction is not required for the present lecture.

Historical sources referred to in Section 5

These are cited for the positions attributed to them and are not required reading.

d'Alembert (1754)	"Croix ou Pile", <i>Encyclopédie</i> — the two-toss counting example
Laplace (1812)	<i>Théorie analytique des probabilités</i> — the classical definition
Venn (1866)	<i>The Logic of Chance</i> — the frequency account
Bertrand (1889)	<i>Calcul des probabilités</i> — the chord example of Section 8
Hilbert (1900)	Sixth problem: an axiomatic treatment of probability
Lebesgue (1902)	Measure and integration
Vitali (1905)	A subset of the line that is not Lebesgue measurable
Borel (1909)	"Les probabilités dénombrables..." — a limit theorem proved by measure
von Mises (1919, 1928)	The collective and the frequency definition
Keynes (1921)	<i>A Treatise on Probability</i> — the logical account; the principle of indifference
Wiener (1923)	Brownian motion as a measure on a function space
Ramsey (1926)	"Truth and Probability" — coherence and betting quotients
Fisher (1930)	"Inverse probability" — the fiducial argument
de Finetti (1931, 1937)	Subjective probability; the exchangeability representation theorem
Kolmogorov (1933)	<i>Grundbegriffe der Wahrscheinlichkeitsrechnung</i> — the axioms
Jeffreys (1939), Carnap (1950)	Later developments of the logical account
Solovay (1970)	Consistency of "every set of reals is Lebesgue measurable" with ZF + DC